

Use AWS/GCP/Any Other Public Cloud Service to Create a VM to Leverage Auto Scaling and Security

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Course: Virtualization and Cloud Computing

Professor: Sumit Kalra

1.Objective –

Set up a virtual machine in a Cloud Platform, implement auto-scaling policies based on workload, and configure security measures like firewall rules and IAM.

2. Cloud Platform Used

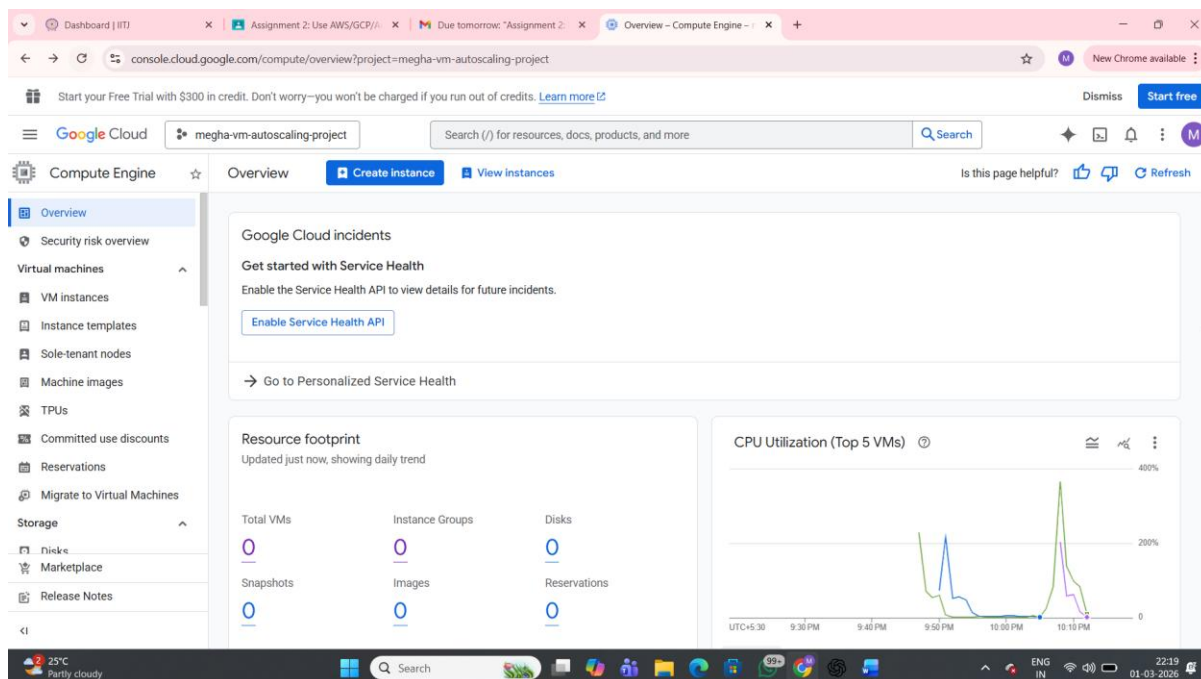
Cloud Provider: Google Cloud Platform (GCP)

Service Used: Compute Engine

Project Name: megha-vm-autoscaling-project

Region: asia-south1 (Mumbai)

Zone: asia-south1-c



3. Step-by-Step Implementation

3.1 Creation of VM Instance (via Instance Template)

Instead of creating a standalone VM, an **Instance Template** was created so that multiple identical VMs can be launched using a Managed Instance Group.

Steps Performed:

1. Navigated to:

Compute Engine → Instance Templates → Create Instance Template

2. Configuration Details:

- Machine Type: **E2-micro**
- Boot Disk: Ubuntu (default)
- Region: asia-south1
- Zone: asia-south1-c

3. Identity & API Access:

- Service Account: Default Compute Engine Service Account
- Access Scope: Allow default access

4. Firewall:

- Enabled: Allow HTTP traffic

Dashboard | ITJ | Assignment 2: Use AWS/GCP/... | Due tomorrow: Assignment 2 | Create an instance template - ...

console.cloud.google.com/compute/instanceTemplates/add?project=megha-vm-autoscaling-project

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Google Cloud | megha-vm-autoscaling-project | Search (/) for resources, docs, products, and more

Compute Engine | Create an instance template

Overview | Security risk overview | Virtual machines | VM instances | Instance templates | Sole-tenant nodes | Machine images | TPUs | Committed use discounts | Reservations | Migrate to Virtual Machines | Storage | Nickle | Marketplace | Release Notes

Describe a VM instance once and then use that template to create groups of identical instances [Learn more](#)

Name *
megha-webserver-template

Manage tags and labels

Location
Global instance templates can be used in any region. To lower the impact of outages outside your region and gain data residency within your region, use a regional instance template. [Learn more](#)

☒ Regional (recommended)
☐ Global

Region *
asia-south1 (Mumbai)

Machine configuration

☒ General purpose | Compute optimized | Memory optimized | Storage optimized | GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	CPU Platform
General purpose				

[Create](#) [Cancel](#) [Equivalent command line](#) [Equivalent REST](#)

You can always create an instance template free of charge. These are estimated costs for a VM instance created using this template:

Monthly estimate
\$8.54
That's about \$0.01 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 1 GB memory	\$7.34
10 GB balanced persistent disk	\$1.20
Logging	Cost varies
Monitoring	Cost varies
Total	\$8.54

[Compute Engine pricing](#)
[Cloud Operations pricing](#)
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25°C Partly cloudy | Search | ENG IN | 22:26 | 01-03-2026

Dashboard | ITI

Assignment 2: Use AWS/GCP/...

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Create an instance template -

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Compute Engine

Create an instance template

Machine configuration

General purpose

Compute optimized

Memory optimized

Storage optimized

GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	CPU Platform
☐ C4	Consistently high performance	2 - 288	4 - 2,232 GB	Intel Emerald Rapids
☐ C4A	Arm-based consistently high performance	1 - 96	2 - 768 GB	Google Axion
☐ C4D	Consistently high performance	2 - 384	3 - 3,072 GB	AMD Turin
☐ N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald Rapids
☐ N4A	Arm-based flexibility & cost optimization	1 - 64	2 - 512 GB	Google Axion
☐ N4D	Flexible & cost-optimized	2 - 96	4 - 768 GB	AMD Turin
☐ C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire Rapids
☐ C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒ E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel Broadwell
☐ N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade Lake
☐ N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD Milan
☐ T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra
☐ T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD Milan

Create

Cancel

Equivalent command line

Equivalent REST

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Due tomorrow: "Assignment 2..."

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Compute Engine

Create an instance template

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

Preset

Custom

e2-micro (2 vCPU, 1 core, 1 GB memory)

vCPU

0.25-2 vCPU (1 shared core)

Memory

1 GB

[Advanced configurations](#)

Provisioning model

VM provisioning model

Standard

Determines how long your VMs will run. Choice is permanent

[VM provisioning model advanced settings](#)

Display device

Enable to use screen capturing and recording tools.

Create

Cancel

Equivalent command line

Equivalent REST

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Compute Engine

Create an instance template

Container

Deploy a container image to this VM instance

[Deploy container](#)

Boot disk

Name	megha-webserver-template
Type	New balanced persistent disk
Size	10 GB
License type	Free
Image	Ubuntu 22.04 LTS

[Change](#)

Identity and API access

Service accounts

Service account

720653890986-compute@developer.gserviceaccount.com

To access instances with this service account you need to add the Service Account User role: roles/iam.serviceAccountUser. [Learn more](#)

Create

Cancel

Equivalent command line

Equivalent REST

You can always create an instance template free of charge. These are estimated costs for a VM instance created using this template:

Monthly estimate

\$8.54

That's about \$0.01 hourly

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Total	\$8.54

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The top screenshot shows the 'Identity and API access' tab in the 'Create an instance template' wizard. It displays the service account '720653890986-compute@developer.gserviceaccount.com' and the 'Allow default access' scope. The 'Firewall' section has 'Allow HTTP traffic' checked. The 'Advanced options' and 'Networking' sections are collapsed. The 'Monthly estimate' on the right shows a total of \$8.54.

The bottom screenshot shows the 'Management' tab. It includes a 'Description' field, 'Reservations' (set to 'Use automatic selection'), and an 'Automation' section with a startup script: `apt install -y apache2`, `echo "Hello from Megha Auto Scaling VM" > /var/www/html/index.html`, and `systemctl restart apache2`. The 'Monthly estimate' remains at \$8.54.

Startup Script

```
#!/bin/bash
apt update
apt install -y apache2
echo "Hello from Megha Auto Scaling VM" > /var/www/html/index.html
systemctl restart apache2
```

3.2 Creation of Managed Instance Group (MIG)

A Managed Instance Group was created to enable auto-scaling and centralized management of VM instances.

Steps:

1. Navigated to:
Compute Engine → Instance Groups → Create Instance Group
2. Selected:
 - Group Type: Managed

- Location: Single Zone
- Zone: asia-south1-c
- Instance Template: megha-webserver-template

3. Initial Target Size:

- 1 Instance

Dashboard | ITJ x Assignment 2: Use AWS/GCP/... x Due tomorrow: "Assignment 2" x Create Instance Group - Comp... x

console.cloud.google.com/compute/instanceGroups/add?project=megha-vm-autoscaling-project

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Compute Engine / Instance groups / Create instance group

Create Instance Group

[Equivalent code](#) [Learn](#)

New managed instance group (stateless)
Automatically manage groups of VMs that do stateless serving and batch processing.

New managed instance group (stateful)
Automatically manage groups of VMs that have persistent data or configurations (such as databases or legacy applications).

New unmanaged instance group
Manually manage groups of load balancing VMs.

Set up automatic management for a group of stateless VMs, including updates, regional deployments, load balancing, autoscaling, and autohealing. [Learn more](#)

Name *
megha-webserver-group ⓘ
Name is permanent

Description

Instance template *
megha-webserver-template ⓘ
e2-micro, ubuntu-2204-jammy-v20260226, asia-south1

Number of instances *
1

☐ Use resize request to create VMs all at once
Create and run VMs all at once when resources become available. Use as an alternative to VMs getting created on an individual basis. [Learn more](#)

Instance flexibility
Calent multibla marbina tunas for our instance group. This improves chances of restoration.

[Create](#) [Cancel](#) [Equivalent code](#)

Instance selections
Create groupings of multiple machine types and rank them in order of preference. [Learn more](#)

[Add selections](#)

Location
For higher availability, select multiple zones in a region instead of a single zone. [Learn more](#)

☒ Single zone
☐ Multiple zones

Region *
asia-south1 (Mumbai) ⓘ

Zone *
asia-south1-c ⓘ

Autoscaling
Use autoscaling to automatically add and remove instances to the group for periods of high and low load. [Learn more](#)

[Configure Autoscaling](#)

[Create](#) [Cancel](#) [Equivalent code](#)

Dashboard | ITI

Assignment 2: Use AWS/GCP/...

Due tomorrow: Assignment 2

Instance groups - Compute En...

console.cloud.google.com/compute/instanceGroups/list?project=megha-vm-autoscaling-project

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Compute Engine / Instance groups

OverviewSecurity risk overviewVirtual machinesVM instancesInstance templatesSole-tenant nodesMachine imagesTPUsCommitted use discountsReservationsMigrate to Virtual MachinesStorageDisksMarketplaceRelease Notes

Instance groups

Create instance groupRefreshDelete

Learn

Instance groups are collections of VM instances that use load balancing and automated services, like autoscaling and autohealing. [Learn more](#)

Filter Enter property name or value

Status	Name	Instances	Template	Group type	Creation time	Recommendation	Autoscaling	Location
☐	megha-webserver-group	1	megha-webserver-template (Regional)	Managed	Mar 1, 2026, 10:38:37 PM UTC+05:30		No configuration	asia-south1-c

24°CPartly cloudy

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22:3901-03-2026

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Assignment 2: Use AWS/GCP/...

Due tomorrow: Assignment 2

Compute Engine - megha-vm -

console.cloud.google.com/compute/instanceGroups/details/asia-south1-c/megha-webserver-group?project=megha-vm-autoscaling-project

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Compute Engine / Instance groups / Instance group: megha-webserver-group

OverviewSecurity risk overviewVirtual machinesVM instancesInstance templatesSole-tenant nodesMachine imagesTPUsCommitted use discountsReservationsMigrate to Virtual MachinesStorageDisksMarketplaceRelease Notes

← megha-webserver-group

EditRestart/replace VMsDelete Group

Equivalent code

OverviewDetailsMonitoringErrors

Instances by status1 instance

Instance by healthNot configuredAutohealing off: [Configure](#)

AutoscalingOn (min 1, max 2)Based on 1 metric and 0 schedules

StatusReady

Creation TimeMar 1, 2026, 10:38:37 PM UTC+05:30

Description

Target running size1

Template[megha-webserver-template \(Regional\)](#)

Locationasia-south1-c

Resize requestsNone

VM instances

SuspendStopStart / ResumeRemove from groupDelete

Filter Enter property name or value

Status	Name	Creation Time	Template	Per instance config	Internal IP	External IP	Health Check Stat	Connect
☐	megha-webserver-group-6d24	Mar 1, 2026, 10:38:48 PM UTC+05:30	megha-webserver-template (Regional)		10.160.0.5 (nic0)	34.14.167.241		SSH

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22:3901-03-2026

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Assignment 2: Use AWS/GCP/...

Due tomorrow: Assignment 2

Compute Engine - megha-vm -

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OverviewSecurity risk overviewVirtual machinesVM instancesInstance templatesSole-tenant nodesMachine imagesTPUsCommitted use discountsReservationsMigrate to Virtual MachinesStorageDisksMarketplaceRelease Notes

← megha-webserver-group

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Creation TimeMar 1, 2026, 10:38:37 PM UTC+05:30

Description

Target running size1

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VM instances

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Filter Enter property name or value

Status	Name	Creation Time	Template	Per instance config	Internal IP	External IP	Health Check Stat	Connect
☐	megha-webserver-group-6d24	Mar 1, 2026, 10:38:48 PM UTC+05:30	megha-webserver-template (Regional)		10.160.0.5 (nic0)	34.14.167.241		SSH

3.3 Configuring Auto-Scaling Policy

Auto-scaling was enabled to automatically increase or decrease instances based on CPU utilization.

Configuration:

- Autoscaling Mode: ON
- Minimum Instances: 1
- Maximum Instances: 2
- Scaling Signal: CPU Utilization
- Target CPU Utilization: 60%

This ensures:

- If CPU usage increases beyond threshold → new instance is created.
- If CPU usage decreases → extra instance is removed.

The screenshot displays the Google Cloud Console interface for creating an instance group. The browser tabs at the top include 'Dashboard | ITJ', 'Assignment 2: Use AWS/GCP/...', 'Due tomorrow: Assignment 2:', and 'Create Instance Group - Comp...'. The address bar shows the URL 'console.cloud.google.com/compute/instanceGroups/add?project=megha-vm-autoscaling-project'. The page header includes the Google Cloud logo, the project name 'megha-vm-autoscaling-project', and a search bar. The breadcrumb trail indicates the path: 'Compute Engine / Instance groups / Create instance group'. The main section is titled 'Create Instance Group' and features a 'Create' button, a 'Cancel' button, and a link to 'Equivalent code'. The 'Autoscaling' section is expanded, showing the 'Autoscaling mode' set to 'On: add and remove instances to the group'. Below this, the 'Minimum number of instances' is set to 1 and the 'Maximum number of instances' is set to 2. The 'Autoscaling signals' section is also expanded, showing 'CPU utilization: 60% (default)' with a dropdown arrow and the text 'Predictive autoscaling is off'. There is a link to 'Add a signal'. The 'Autoscaling schedules' section is collapsed. The 'Initialization period' section is expanded, showing a text input field with '60' and a unit dropdown set to 'seconds'. The 'Scale-in controls' section is collapsed. The 'VM instance lifecycle' section is also collapsed. At the bottom of the page, there is a Windows taskbar with various application icons and a system tray showing the time as 22:37 and the date as 01-03-2026.

3.4 Testing Auto-Scaling Using Stress Tool

To simulate workload:

1. SSH into VM instance.
2. Installed stress tool:

```
sudo apt update
```

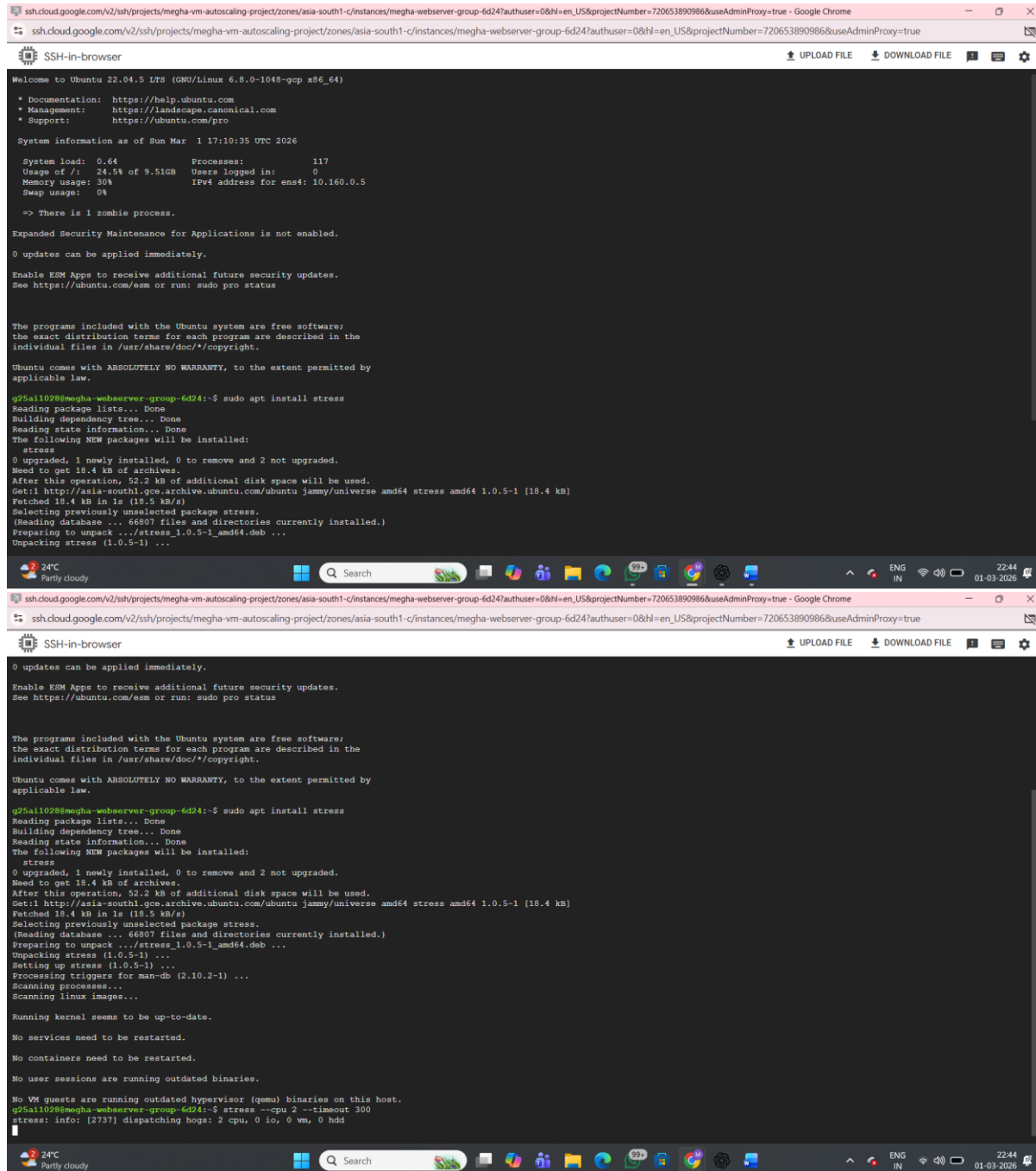
```
sudo apt install stress
```

3. Generated CPU load:

```
stress --cpu 2 --timeout 300
```

Result:

- CPU utilization increased.
- Managed Instance Group automatically scaled from 1 to 2 instances.



The image displays two screenshots of a terminal window accessed via SSH-in-browser. The terminal shows the output of various system commands and the installation of the 'stress' package.

Top Screenshot:

```
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1048-gcp x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sun Mar  1 17:10:35 UTC 2026

System load:  0.64          Processes:           117
Usage of /:   24.5% of 9.51GB Users logged in:       0
Memory usage: 30%          IPv4 address for ens4: 10.160.0.5
Swap usage:   0%

=> There is 1 zombie process.

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

g25a11028@megha-webserver-group-6d24:~$ sudo apt install stress
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://asia-south1.gcp.archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Selecting previously unselected package stress.
(Reading database ... 66807 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
```

Bottom Screenshot:

```
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

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g25a11028@megha-webserver-group-6d24:~$ sudo apt install stress
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  stress
0 upgraded, 1 newly installed, 0 to remove and 2 not upgraded.
Need to get 18.4 kB of archives.
After this operation, 52.2 kB of additional disk space will be used.
Get:1 http://asia-south1.gcp.archive.ubuntu.com/ubuntu jammy/universe amd64 stress amd64 1.0.5-1 [18.4 kB]
Fetched 18.4 kB in 1s (18.5 kB/s)
Selecting previously unselected package stress.
(Reading database ... 66807 files and directories currently installed.)
Preparing to unpack .../stress_1.0.5-1_amd64.deb ...
Unpacking stress (1.0.5-1) ...
Setting up stress (1.0.5-1) ...
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
g25a11028@megha-webserver-group-6d24:~$ stress --cpu 2 --timeout 300
stress: info: [2737] dispatching hogs: 2 cpu, 0 io, 0 vm, 0 hdd
```

Dashboard | ITI x Assignment 2: Use AWS/GCP/... x Due tomorrow: Assignment 2 x Compute Engine - megha-vm- x +

console.cloud.google.com/compute/instanceGroups/details/asia-south1-c/megha-webserver-group?project=megha-vm-autoscaling-project

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Overview Security risk overview Virtual machines VM instances Instance templates Sole-tenant nodes Machine images TPUs Committed use discounts Reservations Migrate to Virtual Machines Storage Disks Marketplace Release Notes

megha-webserver-group Edit Restart/replace VMs Delete Group Equivalent code

The number of instances in the instance group has reached the max_num_replicas. The autoscaler cannot add more instances.

Description Target running size 2 Template megha-webserver-template (Regional) Location asia-south1-c Resize requests None

VM instances Suspend Stop Start / Resume Remove from group Delete

Filter Enter property name or value

Status	Name	Creation Time	Template	Per instance config	Internal IP	External IP	Health Check Stat	Connect
✓	megha-webserver-group-59ht	Mar 1, 2026, 10:41:13 PM UTC+05:30	megha-webserver-template (Regional)		10.160.0.6 (nic0)	34.93.77.195		SSH
✓	megha-webserver-group-6d24	Mar 1, 2026, 10:38:48 PM UTC+05:30	megha-webserver-template (Regional)		10.160.0.5 (nic0)	34.14.167.241		SSH

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console.cloud.google.com/compute/instanceGroups/details/asia-south1-c/megha-webserver-group?project=megha-vm-autoscaling-project

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megha-webserver-group Edit Restart/replace VMs Delete Group Equivalent code

The number of instances in the instance group has reached the max_num_replicas. The autoscaler cannot add more instances.

Overview Details Monitoring Errors

Capacity: 120.00% Utilization: 108.92%

CPU utilization Create alerting policy

Mean utilization: 70.60%

Disk I/O (bytes) Disk I/O (operations)

Logs

Trending videos Minions & More...

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console.cloud.google.com/compute/instanceGroups/details/asia-south1-c/megha-webserver-group?project=megha-vm-autoscaling-project

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megha-webserver-group Edit Restart/replace VMs Delete Group Equivalent code

The number of instances in the instance group has reached the max_num_replicas. The autoscaler cannot add more instances.

Overview Details Monitoring Errors

Instances: 2

Autoscaler utilization (CPU)

Capacity: 120.00% Utilization: 108.92%

CPU utilization Create alerting policy

Logs

4. Security Configuration

A. Firewall Rules Configuration

Firewall rules were configured under:

VPC Network → Firewall

The following rules were verified/created:

1. SSH Rule

- Name: default-allow-ssh
- Protocol: TCP
- Port: 22
- Action: Allow

Purpose:

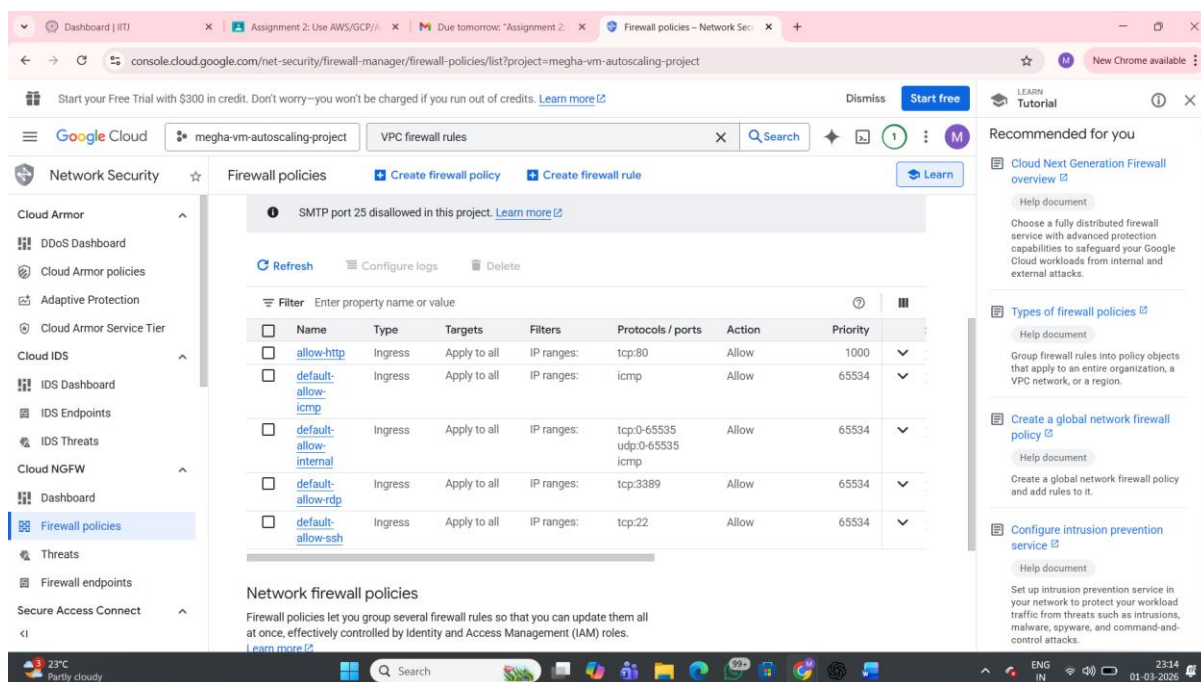
Allows secure remote login to VM instances.

2. HTTP Rule

- Name: allow-http
- Protocol: TCP
- Port: 80
- Action: Allow
- Source: 0.0.0.0/0

Purpose:

Allows web traffic access to VM instances.



B. IAM Roles Configuration

IAM roles were configured to control access permissions.

Navigated to:

IAM & Admin → IAM

Service Account Used:

720653890986-compute@developer.gserviceaccount.com

Assigned Role:

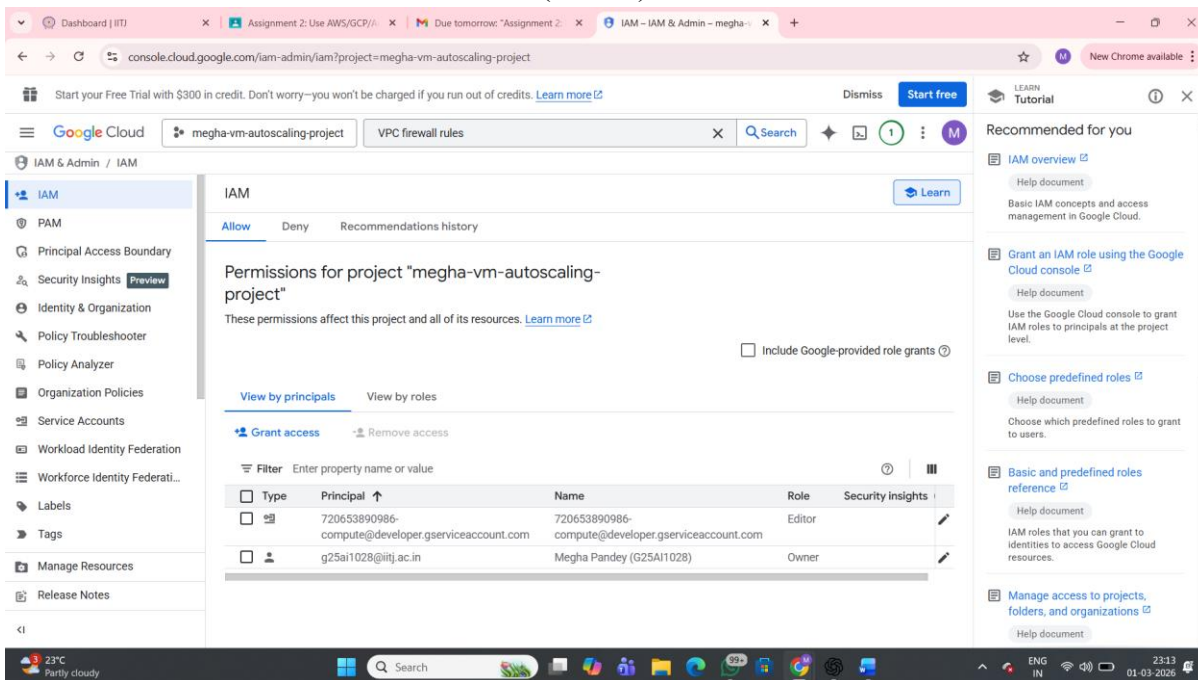
- Editor

User Account:

- Megha Pandey – Owner

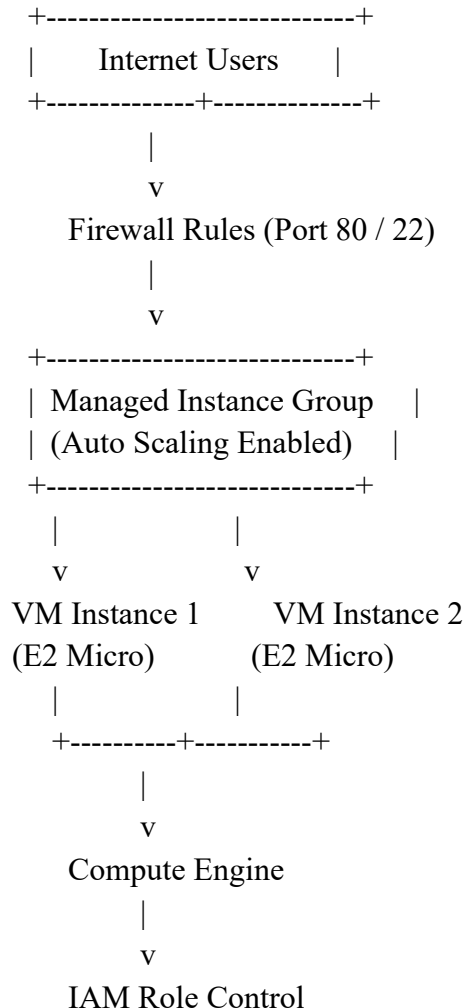
Purpose:

- Restricts and manages access to project resources.
- Ensures role-based access control (RBAC).



5. Architecture Design

Below is the logical architecture of the implementation:



Auto-scaling monitors CPU usage and dynamically adds/removes instances.

6.Results

- VM successfully deployed.
- Managed Instance Group created.
- Auto-scaling successfully triggered using CPU load.
- Firewall rules configured (SSH & HTTP).
- IAM roles assigned and verified.

The system is scalable, secure, and cloud-native.

7. Video Link

<https://drive.google.com/file/d/1-73fXSwy0T-qluRRqxmD3DWy8bTEhjNY/view?usp=sharing>

8.Source Code Repository

GitHub Repository Link:

https://github.com/Codewith-Megha/GCP_VM-to-Leverage-Auto-Scaling-and-Security